System Design Workshop

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IoT

Telco

Linkyfi

GoodLood

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Program stażowy

https://bit.ly/3NNMUYQ

Scala

MongoDB

Angular

Kotlin

Java

Kafka

Kubernetes

Google Cloud Platform

Agenda spotkania

1. Wstęp
2. Wstęp teoretyczny
3. Zadanie - praca w grupach
4. Przerwa (20 min)
5. Zadanie - praca w grupach c.d.
6. Zadanie - dyskusja
7. Feedback i losowanie nagród

Po co się tutaj zebraliśmy?

What is System Design?

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Distributed Systems

- reliability (will it always work?)
- scalability (can it adjust to throughput?)
- availability (what is the uptime?)
- maintainability (logs, metrics, debugging)

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Patterns and tradeoffs

- requirements can be solved with patterns
- not all patterns are correct for every use case
- always justify the tradeoff you are making

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Reliability

- fault tolerant (Prevent downtime)
- no single point of failure (Prevent system collapse)
- preventing cascading failures (Prevent domino-effect)

Ways to achieve reliability

Rate limiter

Trade-off: Hard to find the limit

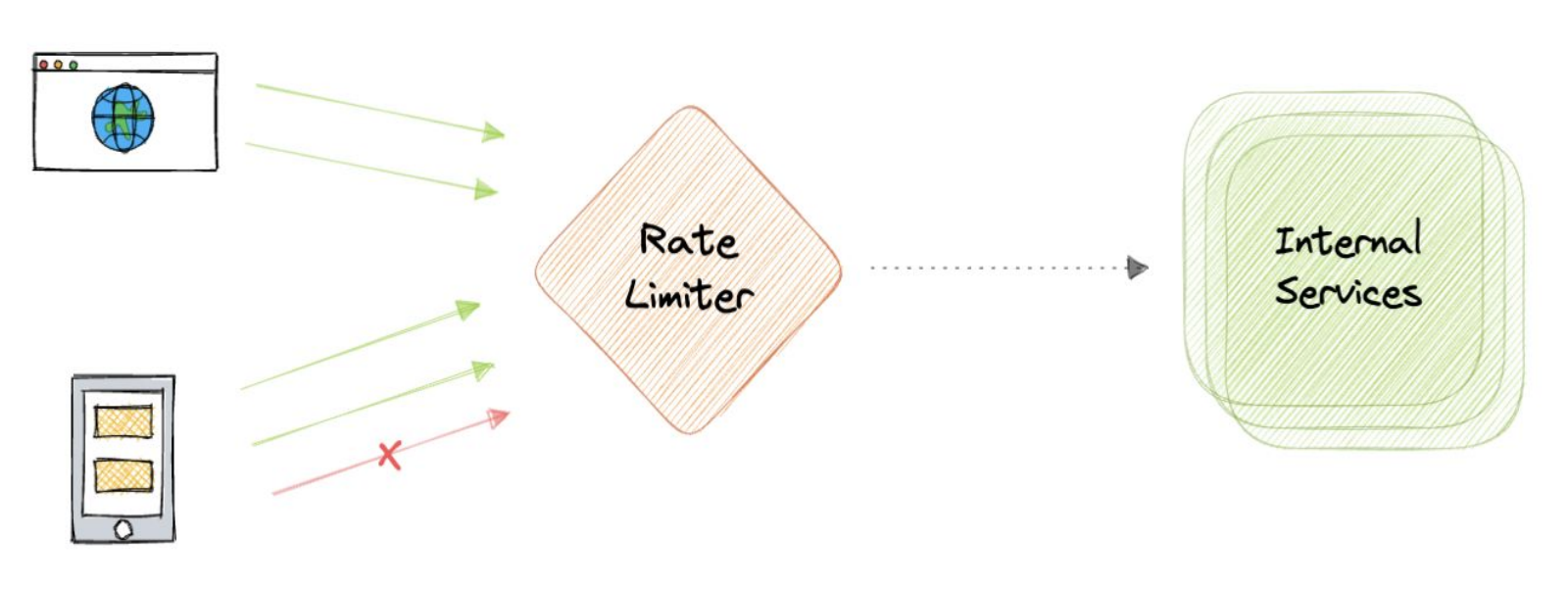


image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#rate-limiting

Message queue

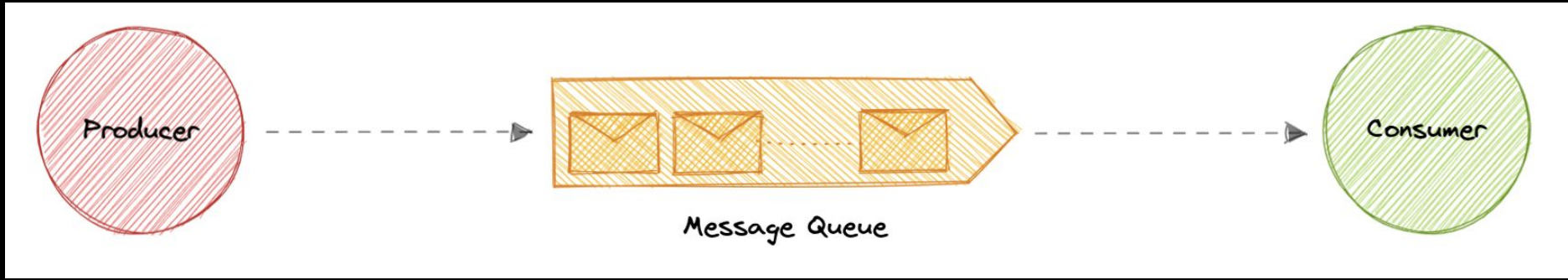


image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#message-queues

Scalability

- scaling up and out
- sharding

Ways to achieve scalability

Message queue

Trade-off: Asynchronous communication

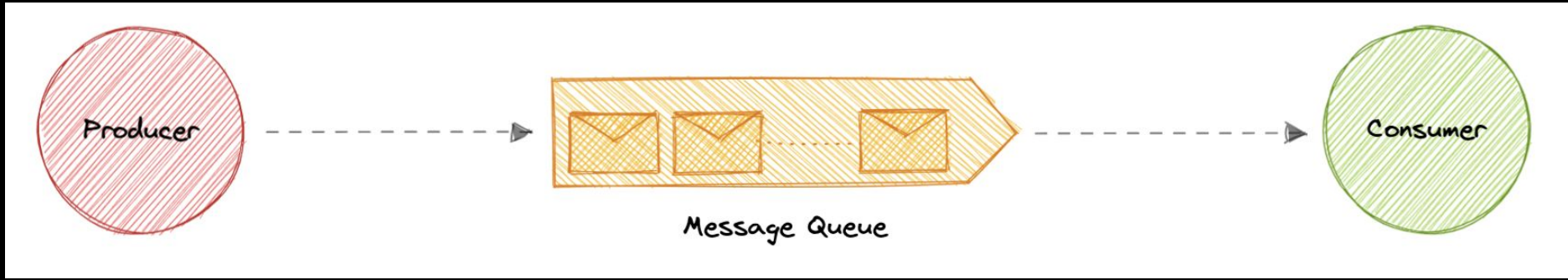


image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#message-queues

Load balancing

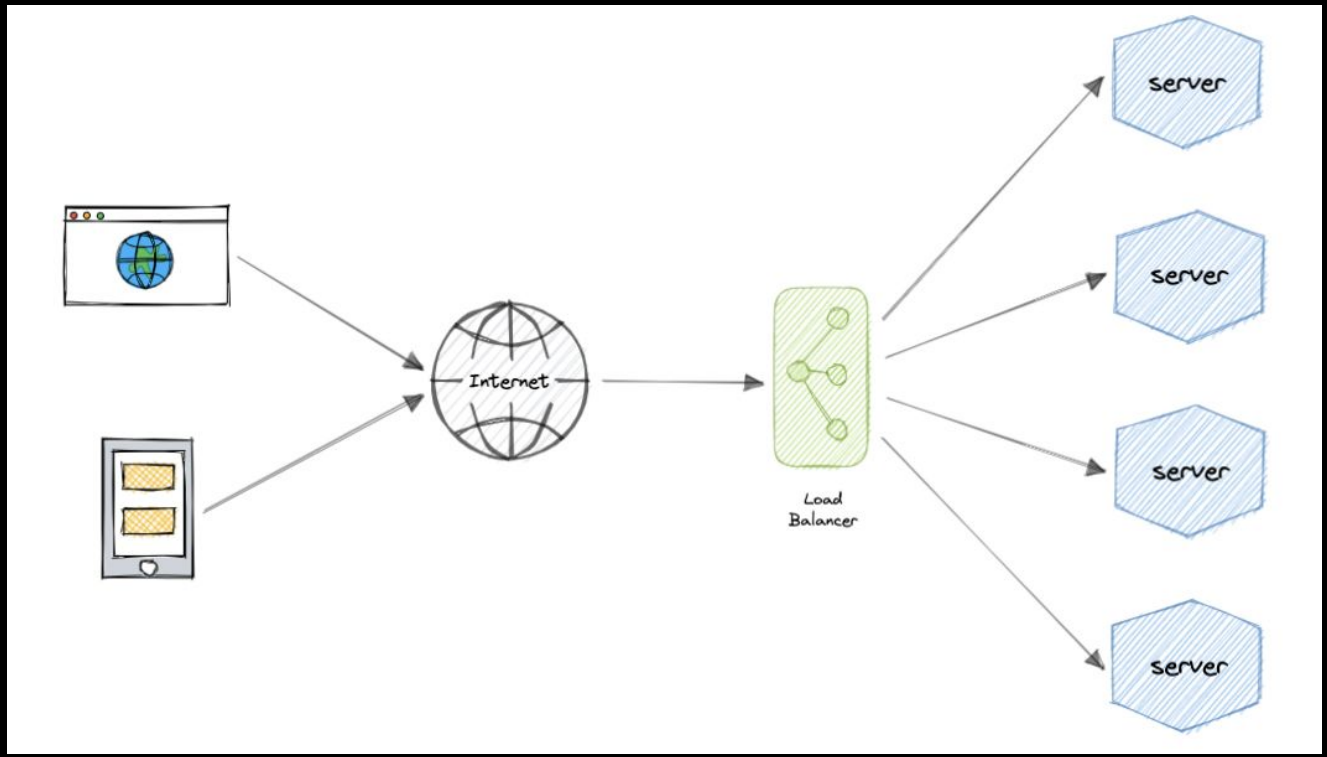
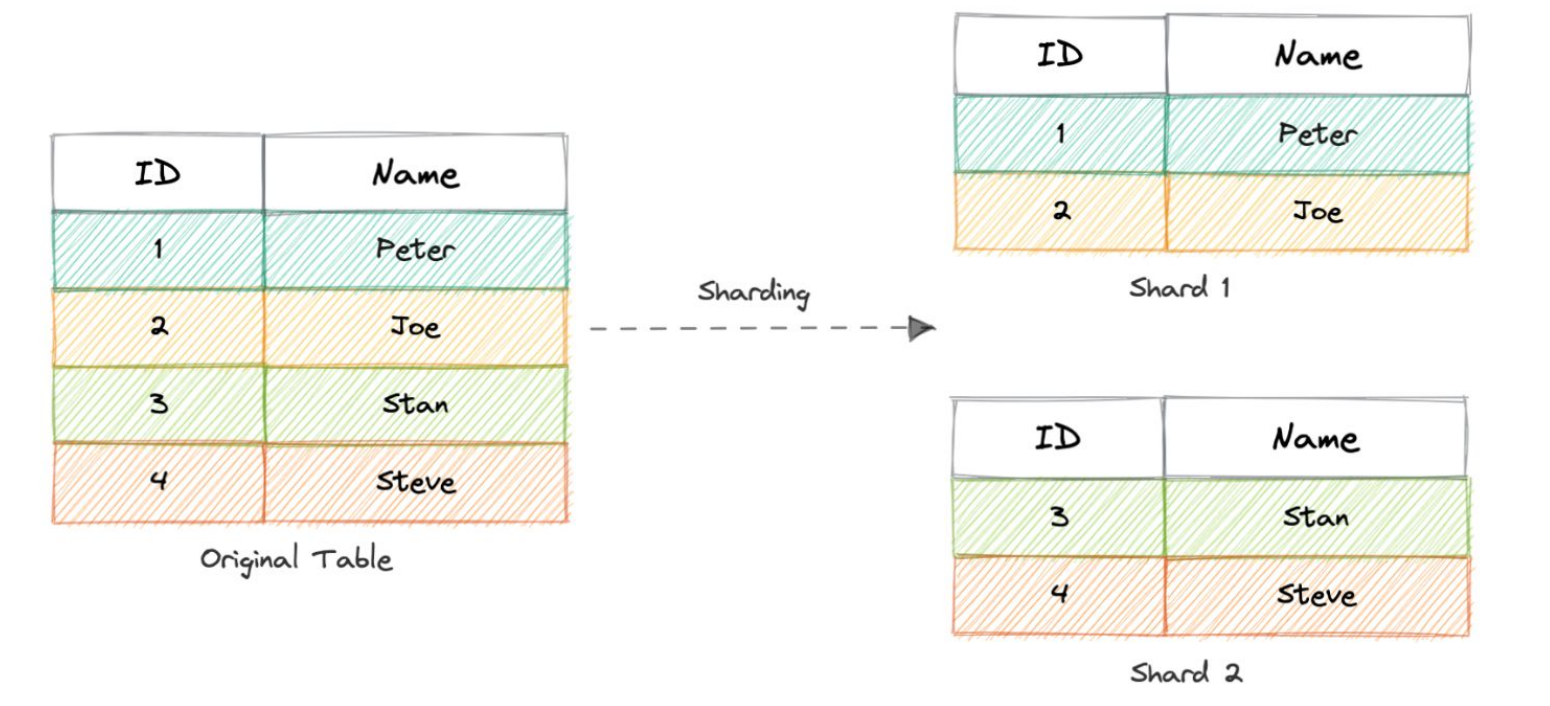


image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#load-balancing

Database sharding



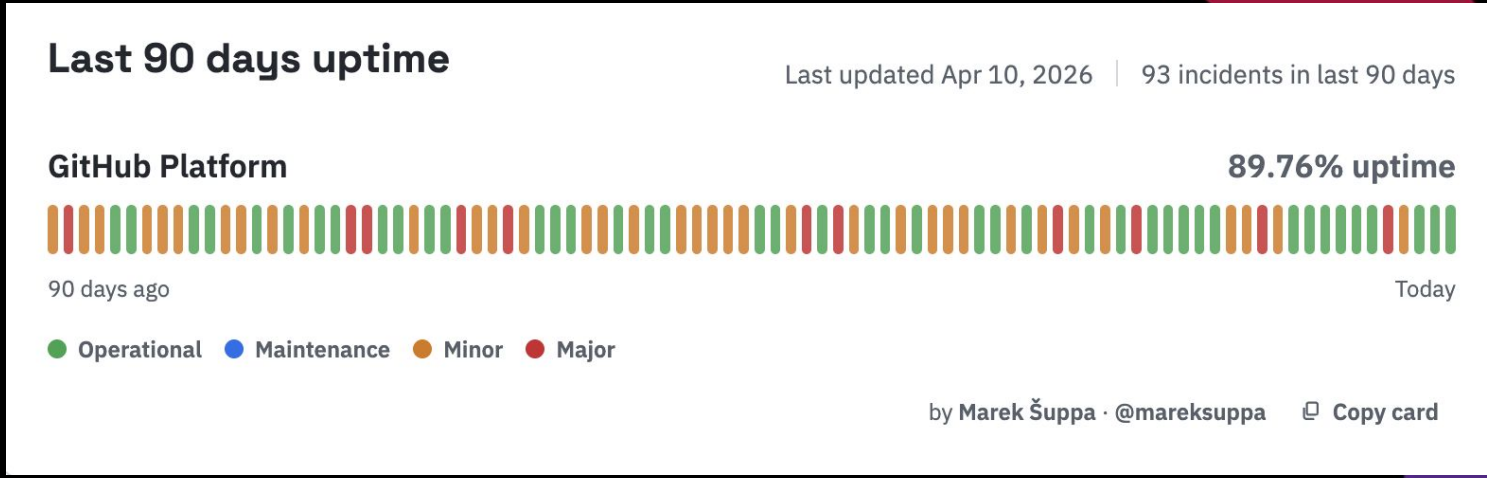
Trade-off:
Unique constraint between shards

image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#what-is-sharding

(High) Availability

- Uptime
- SLA (Service Level Agreement)

Availability in numbers



Availability SLA	Daily	Weekly	Monthly	Yearly
95%	1h	8h	1d	18d
99%	15m	2h	7h	3d
99.9%	1m	10m	40m	9h
99.99%	10s	1m 0.48s	4m	50m
99.999%	1s	6s	40s	5m
99.9999%	0.1s	0.6s	4s	30s

Ways to achieve availability

Fail-over

Trade-off:
Cost increase

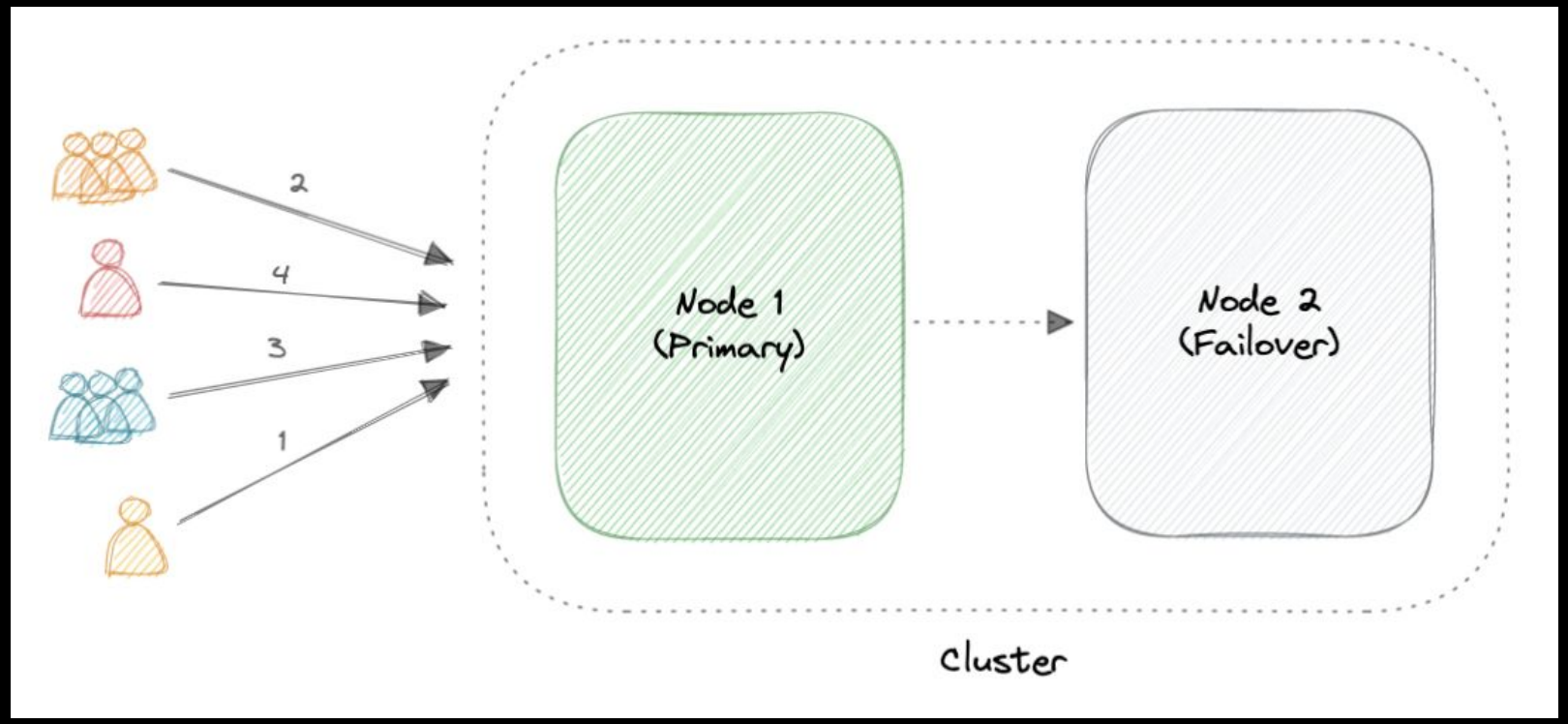


image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#active-passive

Replication

Trade-off:
Increased write latency

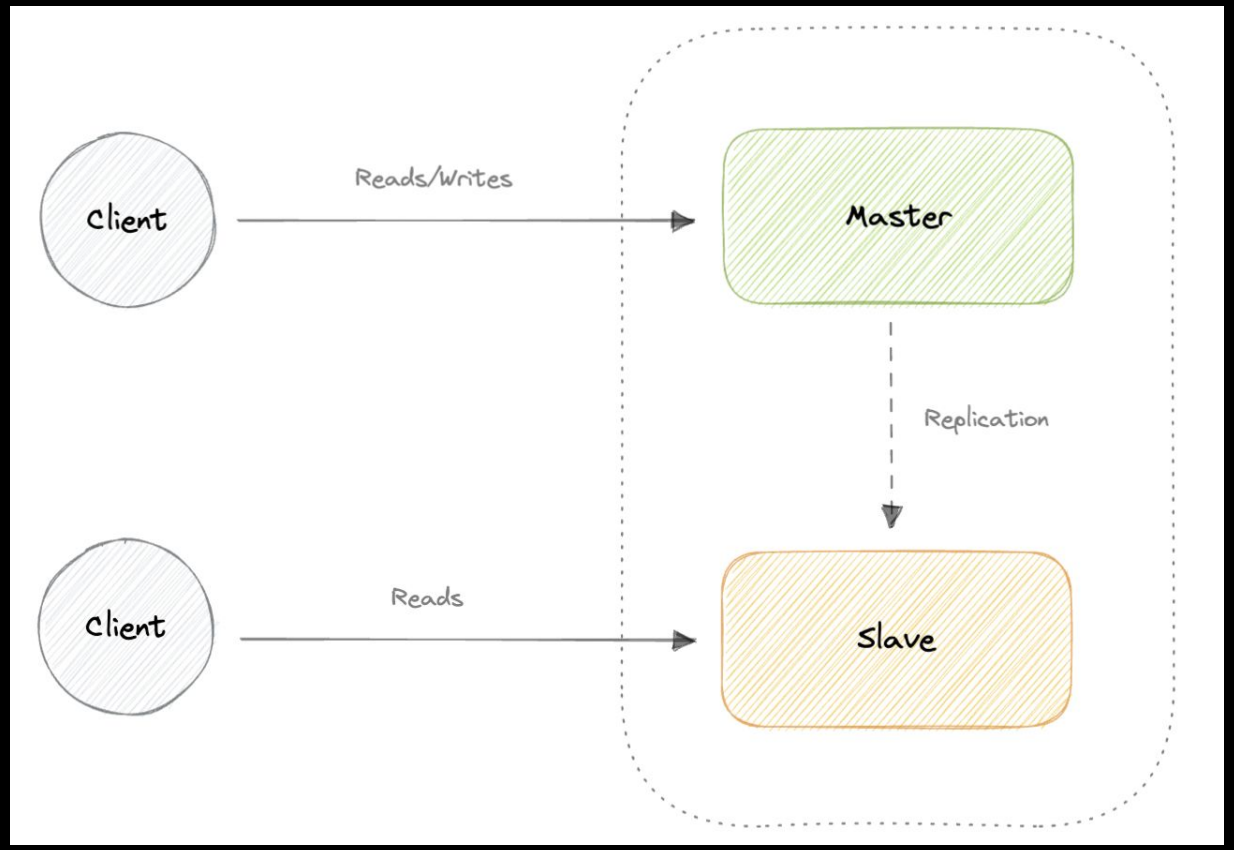


image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#active-passive

Caching

Trade-offs:
Data inconsistencies

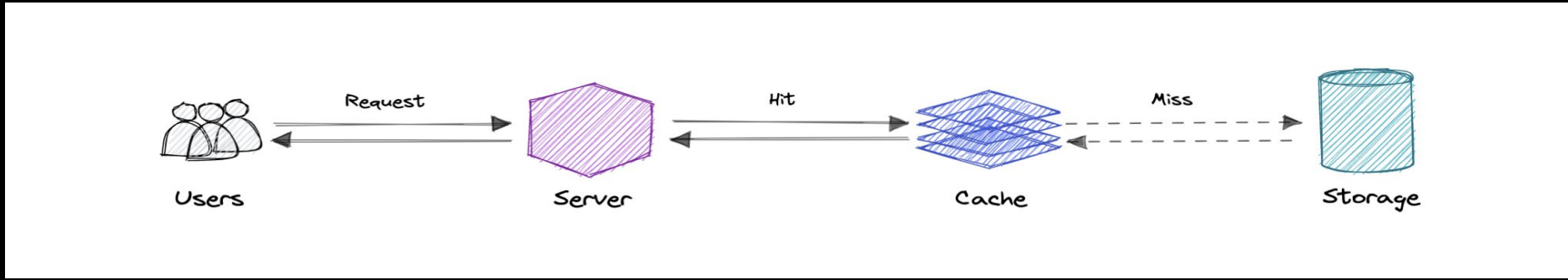


image source: https://github.com/karanpratapsingh/system-design?tab=readme-ov-file#caching

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Maintainability

- simple to understand
- simple to change
- simple to debug

Ways to achieve maintainability

Logging

Logs


```
> {"_entry":"log text  [66027310]","counter":"30057","float":"50.815","wave":-0.8090169943755303,"label":"val1","level":"warning"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [477948175]","counter":"30056","float":"NaN","wave":-0.9510565162949643,"label":"val1","level":"error"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [675786544]","counter":"30055","float":"22.731","wave":-1,"label":"val3","level":"info"}
> {"_entry":"log text  [200456453]","counter":"30054","float":"32.5","wave":-0.95105651629521,"label":"val3","level":"info"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [263531555]","counter":"30053","float":"69.062","wave":-0.8090169943749285,"label":"val2","level":"info"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [864629041]","counter":"30052","float":"5.684","wave":-0.5877852522922731,"label":"val3","level":"info"}
> {"_entry":"log text  [602229084]","counter":"30051","float":"31.989","wave":-0.30901699437450786,"label":"val2","level":"debug"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [841084448]","counter":"30050","float":"NaN","wave":6.770924531994232e-13,"label":"val2","level":"info"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [965740311]","counter":"30049","float":"98.017","wave":0.3090169943757958,"label":"val2","level":"debug"}
> {"_entry":"log text  [239358624]","counter":"30048","float":"4.345","wave":0.587785252291897,"label":"val2","level":"info"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [608925769]","counter":"30047","float":"45.697","wave":0.8090169943757244,"label":"val3","level":"info"}
> {"_entry":"log text with ANSI \u001b[31mpart of the text\u001b[0m [278271372]","counter":"30046","float":"89.092","wave":0.9510565162950664,"label":"val1","level":"debug"}
```

Monitoring

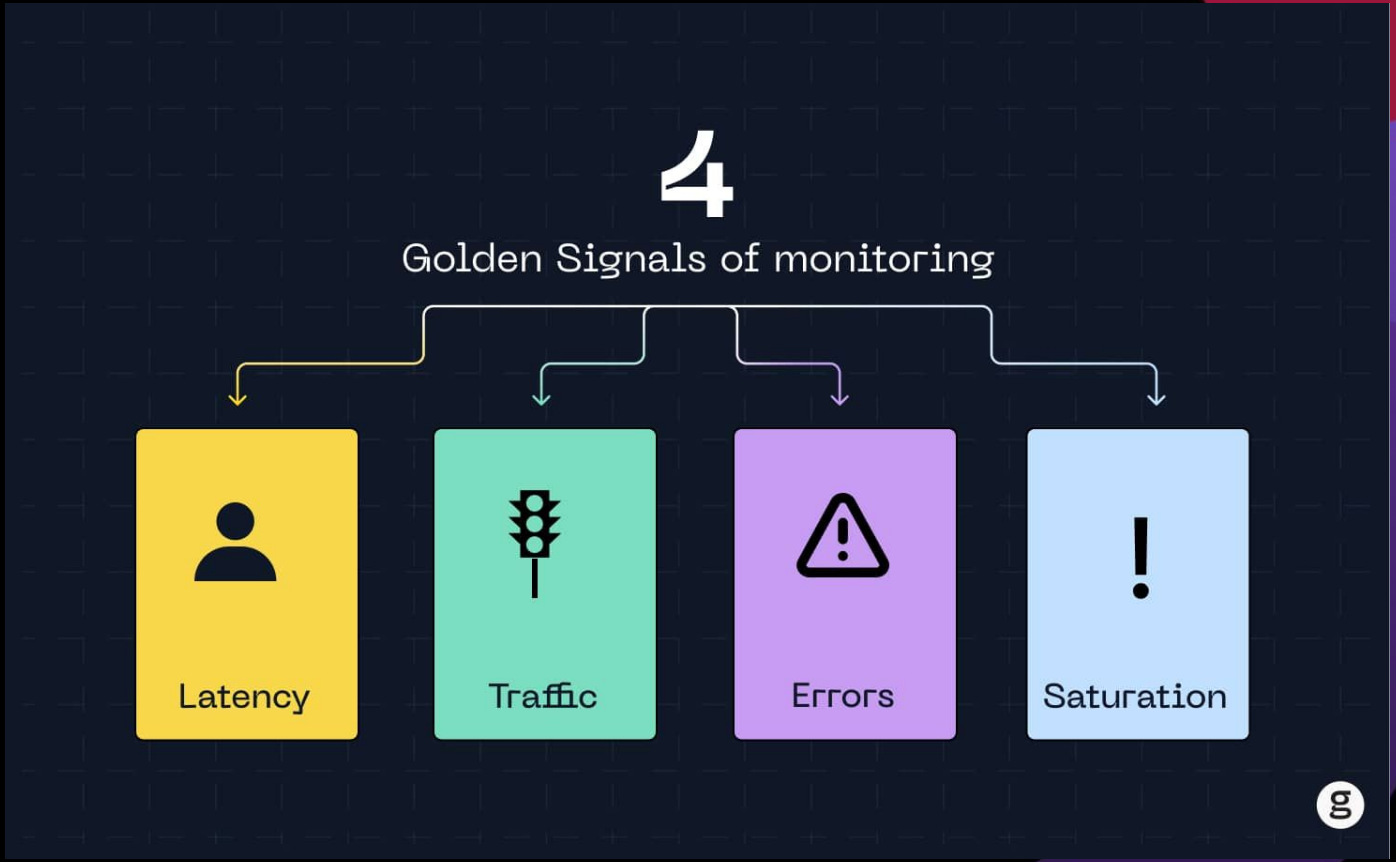

image source: https://www.groundcover.com/blog/4-golden-signals

Agenda rozmowy system design

1. Zbieranie wymagań

Funkcjonalnych:

- co system ma robić?
- czego nie musi robić?

Niefunkcjonalnych:

- jak system ma się zachowywać?
- jak dużo danych / użytkowników?

System - Lista TODO

Funkcjonalne:

Dodawanie i usuwanie rzeczy do zrobienia

Oznaczanie rzeczy jako zrobione

Niefunkcjonalne:

Maksymalnie 10 rzeczy do zrobienia

Czas odpowiedzi <1 sekunda

Średnio 1 request na minutę

Maksymalny rozmiar rzeczy - 1MB (obrazek, tekst, gif), średnio 1KB

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2. Wstępne planowanie

Estymacja zasobów:

- ile danych przechowujemy?
- ile odczytów musimy przetworzyć?
- ile zapisów musimy przetworzyć?

Jakie są główne encje w naszym systemie?

System - Lista TODO

Estymacja zasobów:

- ile danych przechowujemy? <10MB
- ile odczytów musimy przetworzyć? <=1 na minutę
- ile zapisów musimy przetworzyć? <=1 na minutę

Jakie są główne encje w naszym systemie?

TODO (fulfilled, todo_task)

3. Wysokopoziomowy szkic

- główne komponenty
- przepływ danych
- realizacja wymagań funkcjonalnych

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4. Zagłębienie w szczegóły

- wybór bazy danych
- zaprojektowanie API
- zaprojektowanie modeli bazodanowych

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5. Rozszerzenie systemu

- jak możemy spełnić dodatkowe wymaganie X?
- co trzeba zmienić w systemie żeby obsłużyć 10x większy ruch? gdzie są wąskie gardła?
- jak monitorować system, jak znaleźć potencjalne awarie?

Materiały pomocnicze

https://github.com/AVSystem/system-design-workshop-2026

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Feedback

https://bit.ly/4stG5JS

Losowanie nagród

Dziękujemy za uwagę

Przygotowali:
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